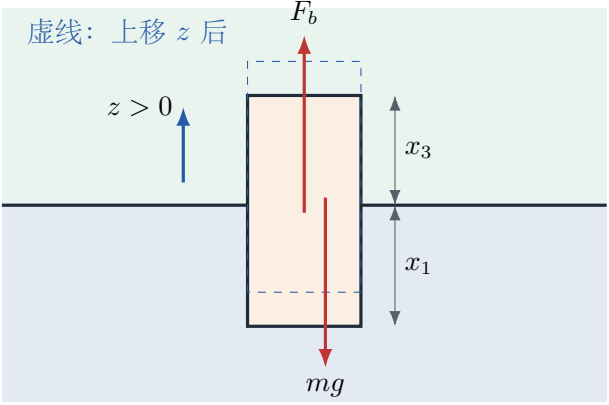

$$\rho_1 x_1 + \rho_3 x_3 = \rho_2 h$$

$$x_3 = h\frac{\rho_1 - \rho_2}{\rho_1 - \rho_3}$$

$$\Delta F_b = -Sg(\rho_1 - \rho_3)z$$